Lab 4 - ELEN

Day 1

- Notes: MEM media instead of DMEM bc ran out

- Step 2:
    - All L929 cells mostly look the same.
    - Confluency: about 40% of the sample plate was covered with cells
    - Morphology: They are a mix of thin and long and round and small cells.
- Step 3: Link to pictures at t=0- here
- Step 4: Materials Used per well:
    - Negative Control: 4 sterilized beads of Cu 3 3/2”
    - Positive Control: 2 sterilized HDPE
    - Ceramic: 2 sterilized 3/26” alumina
    - Metal: 2 sterilized 316L beads of stainless steel
    - Polymer: 6 sterilized ABS chips
    - X: Gloves with 6 cm^2 area
    - Y: KimTech Paper of 6 cm^2 area

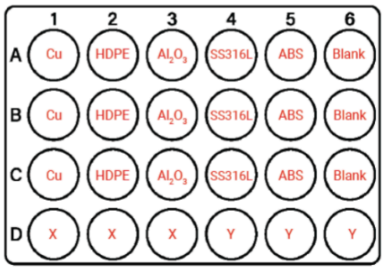


Figure 1 - Schematic of the completed 24-well plate

Table 1 - Observation Chart for Rows and Columns

	1	2	3	4	5	6
A,B,C,D	Cells are long and thin, with a slight bulge in the centre	Cells are a mix of thin and long and round and circular	Cells are mostly circular and small	Cells are a mix of thin and long and round and circular	Cells are a mix of thin and long and round and circular	Cells are a mix of thin and long and round and circular

Day 2

Observations of 24-well plate after incubation:

- The wells with Copper are dark grey in color and the copper appears dull and darker compared to before being added to the cell media
- The kimtech wipes seemed to have absorbed some of the media
- Images of each well after removing material with tweezers - here
- Morphology - wells appear to have a mix of cells with oblong/triangular shapes and with spherical shapes
- Confluency - most of the wells appear to be 70-90% confluent, with a few exceptions that are only 20% confluent. Even wells with high confluency appear to have some spaces that are relatively large in size where no cells are present

96-well plate notes:

- 96 well plate = 7A - 12D
- 7A corresponds to 1A on 48 well plate
- 12D corresponds to 6D on 48 well plate

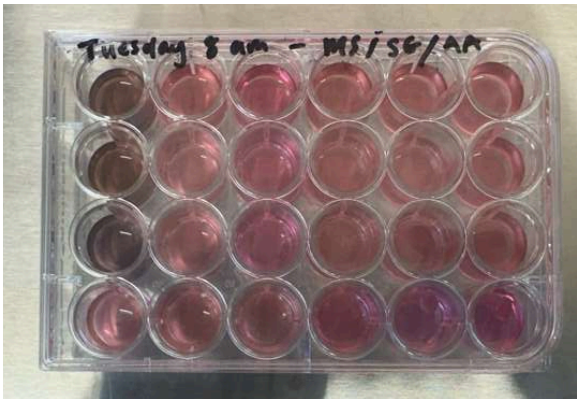


Figure 2 - Image of plate before incubation into 96-well plate

Day 3

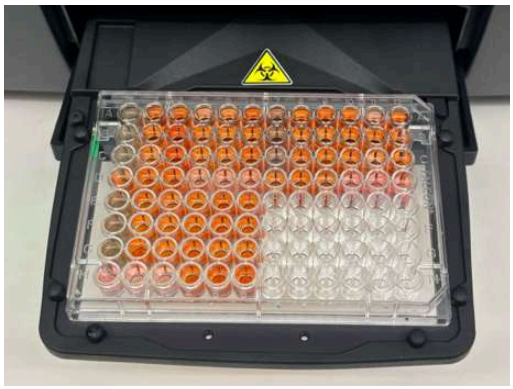


Figure 3 - Image of plate after incubation into 96-well plate

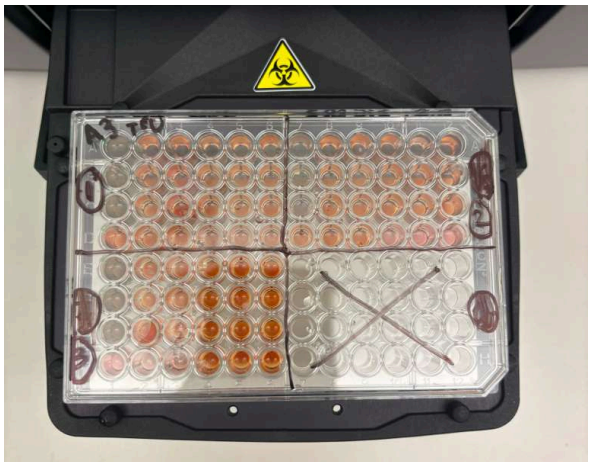


Figure 4 - Image of plate after incubation into 96-well plate with lidl

Evaluation

Direct Contact Test:

Quantitative -

Step 1: Using Figure 1 as a schematic, rank the cytotoxicity of each material.

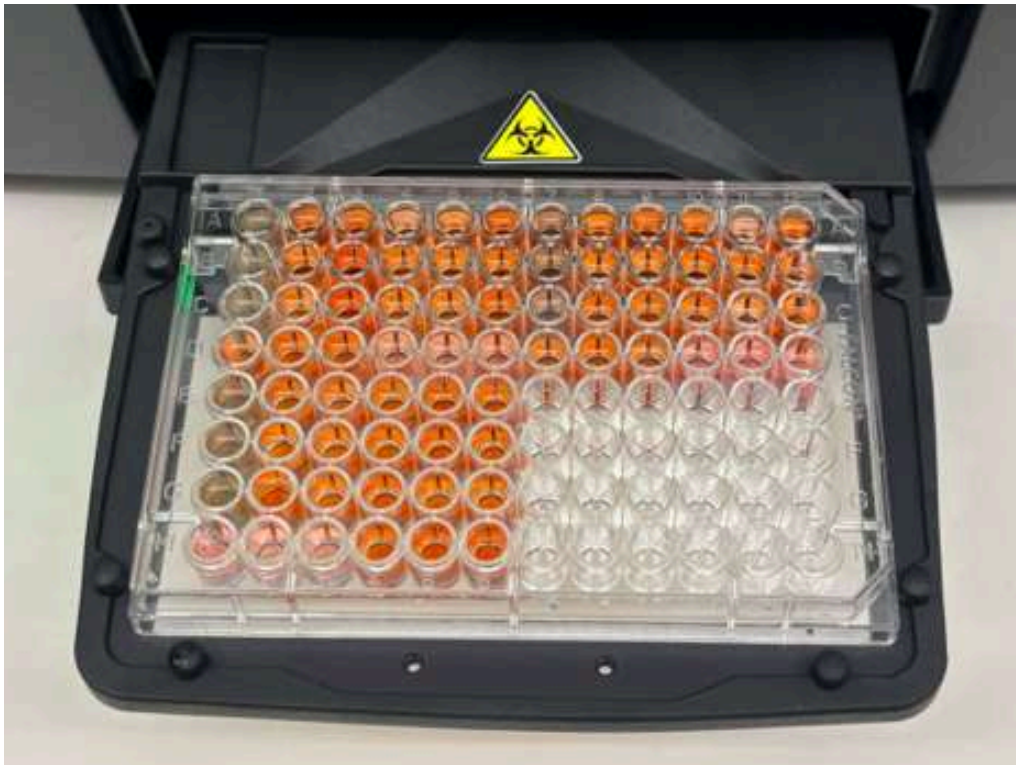


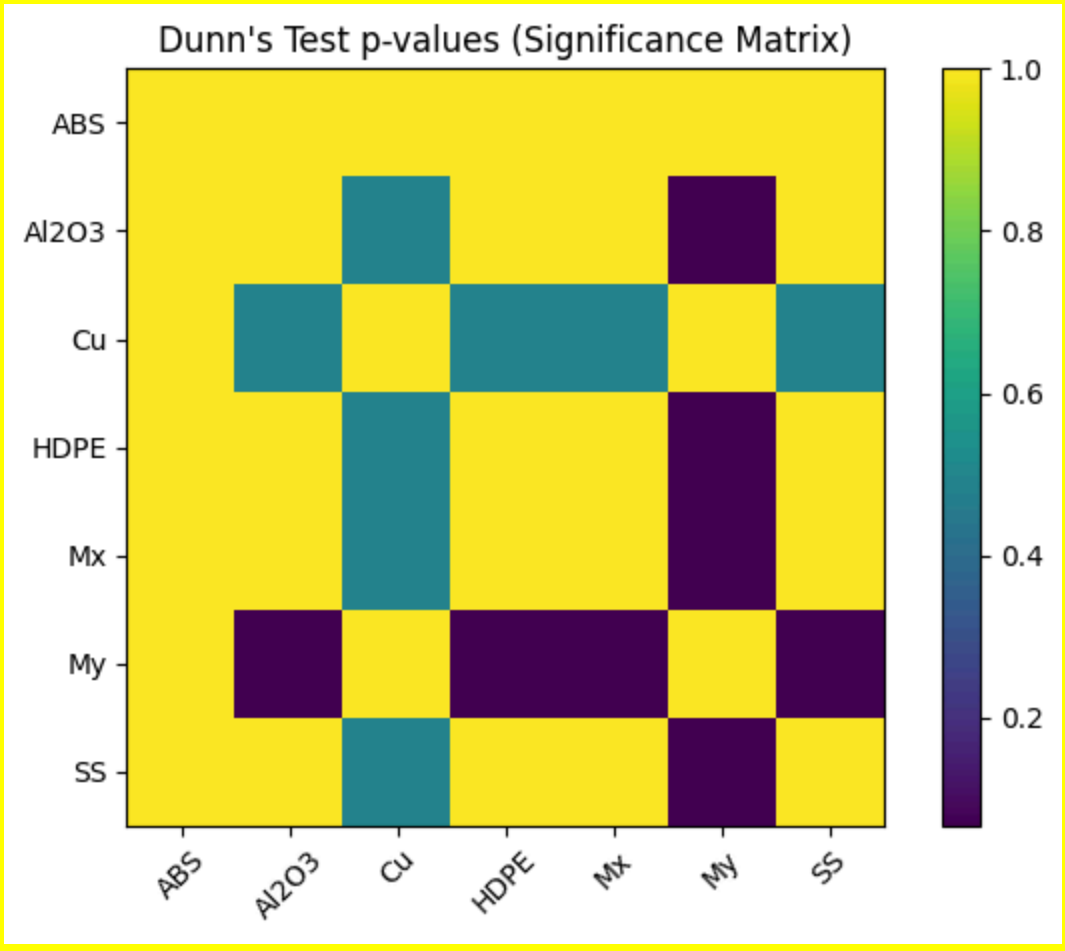
Table 2 - Cytotoxicity Level for Figure 1 schematic where 0 is non-cytotoxic and 3 is severely cytotoxic

	Cu	HDPE	Al2O3	SS	ABS	M x	M y
T1	2	0	0	0	1	0	3
T2	2	0	0	0	0	0	3
T3	2	0	0	0	1	0	3

Step 2: Input into a Python Code to run the Kruskal-Wallis test because there are nominal, more than 2 groups for columns A to C and rows 1 to 6

Code accessible with this link:
https://drive.google.com/file/d/1nr-iamR983bSaG7NHtFZMzJzNX8gzjME/view?usp=sharing

```
Kruskal-Wallis H-statistic: 18.714652956298206
p-value: 0.004673688681712743
Result: Significant differences exist between groups.
```



Quantitative Conclusion

There is no significant difference between the groups A to C and 1 to 6 and X and Y material's cytotoxicity since the shades of pink are not significantly different in ***Figure 2***.

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Extract-Based Contact Test:

Quantitative -

$$Specific\ \ Absorbance\ \ =\ \ [\ Absorbance\ at\ 450\ nm\ \ -\ \ Absorbance\ at\ blank]\ \ -\ \ Absorbance\ at\ 660\ nm$$

Absorbance @ 660 nm	A6, B6, C6	1.674
Absorbance @ 450 nm		
Absorbance @ blank		

XTT assay (Specific Absorbance)

Abs(450)-Blank: 0.0539041666
Abs(650)(test):0.07586666693

	7 - Cu	8-HDPE	9-Al2O3	10-SS316L	11-ABS	Material X	Material Y
Trial 1	1.06762914	0.9459292111	1.411929172	1.164629143	0.4291291651	0.6753291902	0.3290291545
Trial 2	1.196529191	0.8227291522	0.9890292105	1.43502919	0.6655291733	1.391329211	0.4145291862
Trial 3	0.7831291375	1.342729133	1.275429171	1.501329225	0.5139291701	0.9843291459	0.4440291938
Average	1.015762449	1.037129165	1.225462518	1.366995853	0.5361958362	1.016995849	0.3958625115

Ranking of Viability based on Table:

Rank	Material	Average Absorbance	Quantitative Interpretation (Highest to Lowest Viability)
1	SS316L	1.367	Highest Viability
2	Al_2O_3	1.225	High
3	HDPE	1.037	Moderate/High
4	Material X (Gloves)	1.017	Moderate
5	Copper	1.016	Moderate
6	ABS	0.536	Low
7	Material Y (Tissue)	0.396	Low

Statistical Test to Prove Viability:

One Way Anova-
Post-Hoc Test: Tukey - (emphasizes statistical difference between 2 materials)

Stat tests Code:
https://drive.google.com/file/d/1R0_PnPGAHq7nfQreCaQmdPPcSAk8UGFL/view?usp=sharing

=== One-Way ANOVA Results ===
F-statistic: 7.5733
P-value: 0.000913
Conclusion: There is a statistically significant difference in viability between the materials.

=== Tukey HSD Post-Hoc Test Results ===
Multiple Comparison of Means - Tukey HSD, FWER=0.05

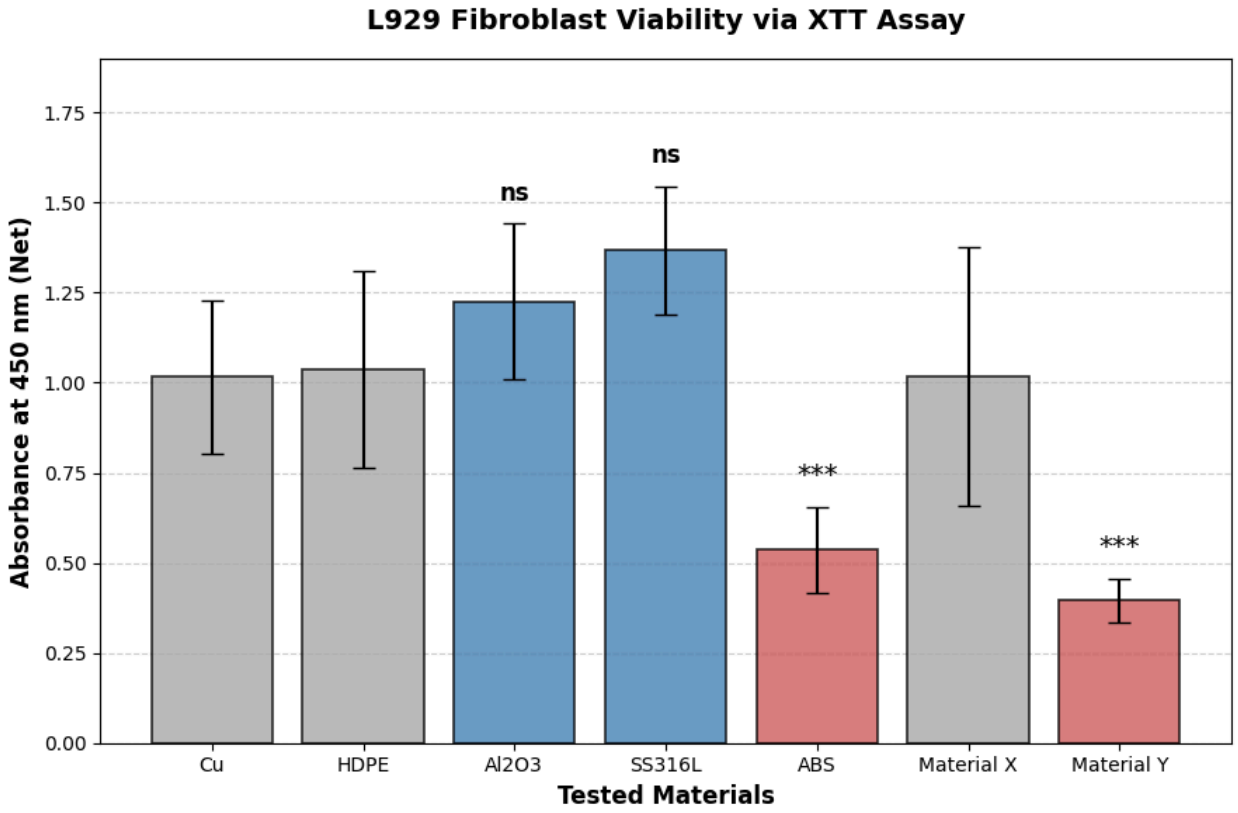
group1	group2	meandiff	p-adj	lower	upper	reject
ABS	Al2O3	0.6893	0.0246	0.0714	1.3072	True
ABS	Cu	0.4796	0.1829	-0.1383	1.0975	False
ABS	HDPE	0.5009	0.1514	-0.117	1.1188	False
ABS	Material X	0.4808	0.1809	-0.1371	1.0987	False
ABS	Material Y	-0.1403	0.9838	-0.7582	0.4776	False
ABS	SS316L	0.8308	0.0059	0.2129	1.4487	True
Al2O3	Cu	-0.2097	0.8983	-0.8276	0.4082	False
Al2O3	HDPE	-0.1883	0.9353	-0.8062	0.4296	False
Al2O3	Material X	-0.2085	0.9007	-0.8264	0.4094	False
Al2O3	Material Y	-0.8296	0.006	-1.4475	-0.2117	True
Al2O3	SS316L	0.1415	0.9831	-0.4764	0.7594	False
Cu	HDPE	0.0214	1.0	-0.5965	0.6393	False
Cu	Material X	0.0012	1.0	-0.6167	0.6191	False
Cu	Material Y	-0.6199	0.049	-1.2378	-0.002	True
Cu	SS316L	0.3512	0.4878	-0.2667	0.9691	False
HDPE	Material X	-0.0201	1.0	-0.638	0.5978	False
HDPE	Material Y	-0.6413	0.0397	-1.2592	-0.0234	True
HDPE	SS316L	0.3299	0.5551	-0.288	0.9478	False
Material X	Material Y	-0.6211	0.0484	-1.239	-0.0032	True
Material X	SS316L	0.35	0.4916	-0.2679	0.9679	False
Material Y	SS316L	0.9711	0.0015	0.3532	1.589	True

True = there is a statistical difference between the materials being compared
Compare materials to Cu and HDPE because they are the controls
When compared to Cu - if false - material and Copper are statistically the same. This is evidence that your material is cytotoxic.

- Note for copper if true but the absorbance value is less than copper that means it is more cytotoxic than copper

When compared to HDPE - if false - good result for biocompatibility. It means the material is statistically the same as the healthy control. This is evidence that the material is viable/non-cytotoxic.

Simplified with barchart below



**** $p < 0.001$ compared to HDPE control; ns = not statistically significant compared to HDPE.*

Chart shows - used based on ANOVA and Tukey test - Al2O3 and SS316L are considered viable while ABS and Material Y are considered cytotoxic

Gray boxes are considered moderate

Red = cytotoxic

Blue = viable

Code for Barchart:

https://drive.google.com/file/d/1KuxxUm1PNu-1P4CW1OKBnBtymMWQnOOd/view?usp=sharing

Lab Report - 4

ELEN: here

1. **You performed the direct contact and extract methods of cytotoxicity per ISO 10993- What are the advantages and disadvantages of performing the direct contact method over the extract method? Remember to cite sources if you research your answer.**

The direct contact method requires less sample preparation and is relatively quicker compared to the extract method. Since the extract method requires several reagents and their preparation, performing the extract method can be tedious. Hence, the direct contact method may be better for applications where cytotoxicity needs to be determined with a smaller quantity of sample and limited resources. A disadvantage of the direct contact method is that the results produced are not quantitative. Instead, results have to be converted to a quantitative from qualitative using the cytotoxicity level chart in the lab manual. Since the interpretation of cytotoxicity levels is based on visually comparing intensity and shades of colors to a guide, interpretation is subjective. This may result in inconsistency when several operators are performing calculations on the same sample and increases risks for experimental error.

2. **Compare your qualitative and quantitative assay results. Do they correlate? Why or why not? What are your evidence-based conclusions about the materials tested?**

The quantitative and qualitative assay results were relatively similar and overall, showed a strong correlation. The materials that resulted in the highest cytotoxicity levels were the same in both test results, and the materials of lower cytotoxicity levels were also relatively similar. More specifically, the materials that exhibited low cytotoxicity scores (of zero) demonstrated higher absorbance values, indicating greater cell viability. Conversely, materials that appeared more cytotoxic visually (scores of 2 or 3) corresponded with lower absorbance values, revealing decreased cell viability.

SS316L, Al_2O_3, and HDPE were scored as non-cytotoxic with scores of 0 across all three of the trials. They also had the highest absorbance values of 1.367, 1.225, and 1.037 respectively. Material X (nitrile gloves in our experiment) also had no visible cytotoxicity and had a relatively high absorbance of about 1.017, indicating increased cell viability and biocompatibility. In contrast, material Y (kimwipes) showed the highest cytotoxicity score of 3 and the lowest absorbance value of 0.396, revealing low cell viability. Copper was a material that had a discrepancy with a low correlation between the qualitative and quantitative results. It had a high cytotoxic score of 2 but had a moderate absorbance that didn't reflect the cytotoxicity score compared to the other materials. This suggests that there might have been some discrepancies between visual appearance in morphology and confluency that didn't reflect a change in metabolic activity of the cells. It could also reveal error between visual qualitative observations and quantitative measurements taken.

In conclusion, SS316L, Al_2O_3, and HDPE are the materials that are the most suitable for applications that involve cell interaction since they resulted in the highest cell viability. Material Y (kimwipes) should be avoided due to having the highest cytotoxic effects while the remaining materials have moderate cytotoxic effects on cells.

3. **Explain the chemistry behind the XTT assay. How does it provide a measure of cell viability, proliferation, and cytotoxicity?**

XTT uses sodium 3´-[1- (phenylaminocarbonyl)- 3,4- tetrazolium]-bis (4-methoxy6-nitro) benzene sulfonic acid hydrate), aka tetrazolium salt, to evaluate cell metabolism by evaluating changes in color. In metabolically active cells, when the mitochondria generates ATP through processes like oxidative phosphorylation and glycolysis, electrons are generated. These electrons are transferred using cofactors (like NADH for example) to mitochondrial dehydrogenase enzymes. These enzymes reduce the XTT reagent, causing it to be converted to a formazan product that is orange in color. The amount of formazan dye is directly proportional to the amount of cellular metabolic activity, also revealing cell viability. This is because only living (viable) cells can reduce the XTT. The XTT assay also provides measurements for cell proliferation since as cells proliferate, the number of metabolically active cells increases. Cell proliferation is represented by increased dye intensity and increasing absorbance values over time assuming that each cell's metabolic activity is relatively constant. XTT can also indicate cytotoxicity since cytotoxic materials cause damage to cells and either decrease their metabolism or kill them. In XTT, this is shown by decreased absorbance values and less formazan dye present (less orange color).

4. For the XTT assay, how would you propose converting absorbance values to cell numbers? Be specific.

First begin by creating a serial dilution of cells with known concentrations. For example, seed a well plate with a range of 0, 5000, 10000, 25000, 50000 and 100000 cells per well. This is the standard plate. This standard will be cultured in the same media and incubated for the same time as the experimental samples to ensure a similar metabolic state, helping to ensure the metabolic rate of the standards are the same as the metabolic rate of the test cells. Next, the XTT reagent would be added to the standard plate. The XTT assay would then be perfomed on both well plates and after the required incubation time (2 hours), the absorbance would be measured at 450nm and at 650nm for reference. Then, subtract the average absorbance from the blank wells from all the other readings to get the net absorbance. Plot the net absorbance as the dependent variable and the known cell number as the independent variable and perform a linear regression test on the resulting data points to determine the equation of the line of best fit i.e $y = mx + b$. The absorbance values from the materials will be used to solve for x ($x = \frac{y-b}{m}$) from the calculated slope from the standard plate. This will help convert the raw absorbance values to an estimate of the cell population per well. Overall, it is important to note that this conversion relies on the assumption that the metabolic rate per cell is constant across the population. This standard curve where known cell quantities are correlated with specific absorbance levels can be used to determine the unknown cell numbers.

5. Another common assay for cytotoxicity is the MTT assay. What advantages does XTT over MTT?

XTT overs several advantages over MTT, mainly due to the differences in solubility of their different reaction products. The product of the MTT assay is an insoluble formazan crystal which must be dissolved in organic solvents before measurements can be taken. This requirement does not exist in XTT since it produces a water-soluble formazan dye, making the process simpler and quicker for analysis. It is also cheaper as the additional organic solvents do not need to be purchased [1]. Since XTT assays are faster and simpler to perform, there is reduced risk for experimental errors to occur. For example, the extra step in MTT assays can cause inconsistent or inaccurate readings if there is uneven or incomplete dissolution of the crystals. XTT assays are therefore easier to reproduce and repeat than MTT assays, resulting in greater consistency. The decreased need for other reagents (organic solvents) in XTT also allows the procedure to be safer than MTT since there are less hazardous materials required [1].

References:

[1] Abcam, “XTT assays for cell viability assessment,” *Abcam Knowledge Center*, 2024. [Online]. Available: https://www.abcam.com/en-us/knowledge-center/cell-biology/xtt-assays-for-cell-viability-assessment. Accessed: Mar. 19, 2026.